

Problems for week 8

Solutions

Problem 1. Suppose $\mathbf{u}, \mathbf{v} : \mathbb{R} \rightarrow \mathbb{R}^3$ are differentiable vector functions. Show that:

$$\frac{d}{dt}(\mathbf{u}(t) \cdot \mathbf{v}(t)) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$$

Solution 1. Let $\mathbf{u}(t) = (f_1(t), f_2(t), f_3(t))$ and $\mathbf{v}(t) = (g_1(t), g_2(t), g_3(t))$. Then

$$\mathbf{u}(t) \cdot \mathbf{v}(t) = \sum_{i=1}^3 f_i(t)g_i(t)$$

and so the ordinary product rule gives

$$\begin{aligned} \frac{d}{dt}(\mathbf{u}(t) \cdot \mathbf{v}(t)) &= \sum_{i=1}^3 \frac{d}{dt}(f_i(t)g_i(t)) = \sum_{i=1}^3 (f'_i(t)g_i(t) + f_i(t)g'_i(t)) \\ &= \sum_{i=1}^3 f'_i(t)g_i(t) + \sum_{i=1}^3 f_i(t)g'_i(t) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t). \end{aligned}$$

Problem 2. Suppose $\mathbf{u}, \mathbf{v} : \mathbb{R} \rightarrow \mathbb{R}^3$ are differentiable vector functions. Show that:

$$\frac{d}{dt}(\mathbf{u}(t) \times \mathbf{v}(t)) = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

Solution 2. Let $\mathbf{u}(t) = (f_1(t), f_2(t), f_3(t))$ and $\mathbf{v}(t) = (g_1(t), g_2(t), g_3(t))$. Then

$$\begin{aligned} \mathbf{u}(t) \times \mathbf{v}(t) &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ f_1(t) & f_2(t) & f_3(t) \\ g_1(t) & g_2(t) & g_3(t) \end{vmatrix} \\ &= (f_2(t)g_3(t) - f_3(t)g_2(t))\hat{\mathbf{i}} - (f_1(t)g_3(t) - f_3(t)g_1(t))\hat{\mathbf{j}} + (f_1(t)g_2(t) - f_2(t)g_1(t))\hat{\mathbf{k}} \end{aligned}$$

On the other hand,

$$\begin{aligned} \mathbf{u}'(t) \times \mathbf{v}(t) &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ f'_1(t) & f'_2(t) & f'_3(t) \\ g_1(t) & g_2(t) & g_3(t) \end{vmatrix} \\ &= (f'_2(t)g_3(t) - f'_3(t)g_2(t))\hat{\mathbf{i}} - (f'_1(t)g_3(t) - f'_3(t)g_1(t))\hat{\mathbf{j}} + (f'_1(t)g_2(t) - f'_2(t)g_1(t))\hat{\mathbf{k}} \end{aligned}$$

and

$$\begin{aligned}\mathbf{u}(t) \times \mathbf{v}'(t) &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ f_1(t) & f_2(t) & f_3(t) \\ g_1'(t) & g_2'(t) & g_3'(t) \end{vmatrix} \\ &= (f_2(t)g_3'(t) - f_3(t)g_2'(t))\hat{\mathbf{i}} - (f_1(t)g_3'(t) - f_3(t)g_1'(t))\hat{\mathbf{j}} + (f_1(t)g_2'(t) - f_2(t)g_1'(t))\hat{\mathbf{k}}\end{aligned}$$

The result follows from noting that the usual product rule implies

$$\begin{aligned}\frac{d}{dt}(f_2(t)g_3(t) - f_3(t)g_2(t)) &= f_2'(t)g_3(t) - f_3'(t)g_2(t) + f_2(t)g_3'(t) - f_3(t)g_2'(t), \\ \frac{d}{dt}(f_1(t)g_3(t) - f_3(t)g_1(t)) &= f_1'(t)g_3(t) - f_3'(t)g_1(t) + f_1(t)g_3'(t) - f_3(t)g_1'(t), \\ \frac{d}{dt}(f_1(t)g_2(t) - f_2(t)g_1(t)) &= f_1'(t)g_2(t) - f_2'(t)g_1(t) + f_1(t)g_2'(t) - f_2(t)g_1'(t).\end{aligned}$$

Problem 3. Consider the vector field

$$\mathbf{F} = (x + y^2)\hat{\mathbf{i}} + xz\hat{\mathbf{j}} + (y + z)\hat{\mathbf{k}}.$$

Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r},$$

where the path C is given in terms of its position vector by

$$\mathbf{r}(t) = t^2\hat{\mathbf{i}} + t^3\hat{\mathbf{j}} - 2t\hat{\mathbf{k}}, \quad 0 \leq t \leq 2.$$

Solution 3. If $x = t^2, y = t^3, z = -2t$, then

$$dx = 2t \, dt, \quad dy = 3t^2 \, dt, \quad dz = -2 \, dt$$

Hence

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (x + y^2) \, dx + xz \, dy + (y + z) \, dz \\ &= \int_0^2 (t^2 + t^6)(2t) \, dt + (-2t^3)(3t^2) \, dt + (t^3 - 2t)(-2) \, dt \\ &= \int_0^2 (4t - 6t^5 + 2t^7) \, dt = \left[2t^2 - t^6 + \frac{t^8}{4} \right]_0^2 = 2^3 - 2^6 + \frac{2^8}{4} = 8.\end{aligned}$$

Problem 4. The force exerted by an electric charge at the origin on a charged particle at a point (x, y, z) with position vector $\mathbf{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is $\mathbf{F}(\mathbf{r}) = K\mathbf{r}/|\mathbf{r}|^3$ where K is a constant. Compute the work done as the particle moves along a straight line from $(2, 0, 0)$ to $(2, 1, 5)$.

Solution 4. The line segment from $(2, 0, 0)$ to $(2, 1, 5)$, denoted L , has parametric equations $x = 2, y = t, z = 5t$ for $0 \leq t \leq 1$. Also,

$$\mathbf{F}(\mathbf{r}) = K \frac{\mathbf{r}}{|\mathbf{r}|^3} = K \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}.$$

It follows that the work is given by

$$\begin{aligned} \int_L \mathbf{F} \cdot d\mathbf{r} &= \int_L \frac{x dx + y dy + z dz}{(x^2 + y^2 + z^2)^{3/2}} = K \int_0^1 \frac{(2)(0) dt + (t)(1) dt + (5t)(5) dt}{(4 + t^2 + (5t)^2)^{3/2}} \\ &= K \int_0^1 \frac{26t}{(4 + 26t^2)^{3/2}} dt = K \left[\frac{-1}{(4 + 26t^2)^{1/2}} \right]_0^1 = K \left(\frac{1}{2} - \frac{1}{\sqrt{30}} \right). \end{aligned}$$

Problem 5. Consider the vector field $\mathbf{F} = yz\hat{i} + xz\hat{j} + (xy + 2z)\hat{k}$, and let C be the line segment from $(1, 0, -2)$ to $(4, 6, 3)$.

- Find a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that $\mathbf{F} = \nabla f$. Show your work.
- Use part (a) to evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Solution 5.

- $f_x(x, y, z) = yz$ implies $f(x, y, z) = xyz + g(y, z)$, and so $f_y(x, y, z) = xz + g_y(y, z)$. But $f_y(x, y, z) = xz$, so $g_y(y, z) = 0$, and (integrating with respect to y) $g(y, z) = h(z)$. Thus $f(x, y, z) = xyz + h(z)$, and $f_z(x, y, z) = xy + h'(z)$. But $f_z(x, y, z) = xy + 2z$, so $h'(z) = 2z$, and hence $h(z) = z^2 + c$. Thus (taking $c = 0$), one potential function is $f(x, y, z) = xyz + z^2$.
- By the fundamental theorem of calculus for line integrals,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(4, 6, 3) - f(1, 0, -2) = 81 - 4 = 77.$$

Problem 6. Consider the vector field $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined as follows:

$$\mathbf{F}(x, y) = \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}.$$

- Show that $\mathbf{F}$ is the gradient of the polar angle function $\theta(x, y) = \tan^{-1}(\frac{y}{x})$ defined over the right half-plane $x > 0$. (Note: this formula for θ does not make sense for $x = 0$!)

- (b) Suppose that C is a smooth planar curve in the right half-plane $\{(x, y) \in \mathbb{R}^2 : x > 0\}$ joining two points $A = (x_1, y_1)$ and $B = (x_2, y_2)$. Express the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

in terms of the polar coordinates (r_1, θ_1) and (r_2, θ_2) of A and B .

- (c) Compute directly from the definition the line integrals

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \text{ and } \int_{C_2} \mathbf{F} \cdot d\mathbf{r},$$

where C_1 is the upper half of the unit circle running from $(1, 0)$ to $(-1, 0)$, and C_2 is the lower half of the unit circle, also going from $(1, 0)$ to $(-1, 0)$.

- (d) Using the results of parts (a)–(c), is the vector field $\mathbf{F}$ conservative over its entire domain of definition? Is it conservative over the right half-plane $x > 0$? Justify your answers.

Solution 6.

- (a) For $\theta(x, y) = \tan^{-1}(\frac{y}{x})$,

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + (y/x)^2} \frac{-y}{x^2} = -\frac{y}{x^2 + y^2}, \text{ and } \frac{\partial \theta}{\partial y} = \frac{1}{1 + (y/x)^2} \frac{1}{x} = \frac{x}{x^2 + y^2}.$$

So $\nabla \theta = \mathbf{F}$.

- (b) Because $\theta(x, y) = \tan^{-1}(\frac{y}{x})$ is well-defined in the right half-plane ($x > 0$) and $\mathbf{F} = \nabla \theta$, the Fundamental Theorem of Calculus for Line Integrals implies

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \theta(x_2, y_2) - \theta(x_1, y_1) = \theta_2 - \theta_1.$$

- (c) Compute:

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \frac{-y dx + x dy}{x^2 + y^2} = \int_0^\pi \frac{(-\sin \theta)(-\sin \theta) + \cos \theta \cos \theta}{\cos^2 \theta + \sin^2 \theta} d\theta = \int_0^\pi d\theta = \pi.$$

Similarly,

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^{-\pi} d\theta = -\int_{-\pi}^0 d\theta = -\pi.$$

Or geometrically: $\text{length}(C_1) = \text{length}(C_2) = \pi$, whereas $\mathbf{F} \cdot \hat{T} = 1$ on C_1 and $\mathbf{F} \cdot \hat{T} = -1$ on C_2 .

- (d) The vector field $\mathbf{F}$ is defined everywhere except at the origin, but is not conservative over its entire domain of definition. Indeed, the two line integrals computed in (c) both run from $(1, 0)$ to $(-1, 0)$, but they are not equal, so path-independence fails. On the other hand, $\mathbf{F}$ is conservative over the half-plane $x > 0$, where $\mathbf{F} = \nabla \theta$ and the Fundamental Theorem of Calculus for Line Integrals gives a formula for the line integral involving only the values of θ at the endpoints (as seen in (b)).